

**Q1. (8 marks)** Let  $V = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 + 2x_2 + x_3 - x_4 = 0\}$  with inner product  $\langle u, v \rangle = 2u_1v_1 + u_2v_2 + u_3v_3 + 2u_4v_4$ . Show that  $V$  is a subspace of  $\mathbb{R}^4$ ; find a basis and its dimension; using the given inner product, apply the Gram-Schmidt process to obtain an orthonormal basis for  $V$ .

**Q2. (7 marks)** Let  $P_2$  denote the vector space of real polynomials of degree  $\leq 2$ . Define  $T : P_2 \rightarrow P_2$  by  $T(p)(x) = p'(x) + p(0)$ . Using the ordered basis  $\{1, x, x^2\}$ , find the matrix of  $T$ , determine  $\text{Null}(T)$  and  $\text{Range}(T)$  with their bases and dimensions, and verify the Rank-Nullity Theorem.

**Q3. (5 marks)** Consider  $A = \begin{pmatrix} 4 & 1 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ . Find the eigenvalues and their algebraic and geometric multiplicities, determine whether  $A$  is diagonalizable, and state the eigenvalues of  $A^2 + 2I$ .

---